Energy from Fluctuations

Mieczysław Barancewicz

26 Marzec 2026

© 2025 M. Barancewicz. All rights reserved. Reproduction and distribution are permitted, provided that original authorship is acknowledged. For the truth has no fear of dissemination or debate.

Abstract

This chapter sheds new light on the origin of particles with extreme kinetic energies ($> 10^{20}$ eV). Utilizing the **Momentum-less Photon (MLP)** paradigm, I demonstrate that acceleration is not the result of processes in distant extragalactic sources, but rather a local statistical phenomenon. This mechanism, analogous to relativistic Brownian motion, is based on the absorption of energy from massive field density fluctuations, where the high-speed particle acts as a resonator.

Mieczysław Barancewicz.
proton4@poczta.pl
07-07-2025
Prawa zastrzeżone.
Nie wolno przypisywać sobie autorstwa.

Contents

1 Astrophysical Implications of the MRG-MLP Model: Resolving the UHECR Paradox **3**
1.1 Critique of the Ballistic Model and the GZK Limit 3
1.2 The Mechanism of Local Field "Surfing" 3
1.3 The Statistical Singularity of the Event 3
1.4 Abrupt Decoherence: Explosion of Accumulated Potential . . 4
1.5 Relativistic Boost Effect 4
1.6 Statistical Selectivity of the Acceleration Mechanism 5

2 Statistical Analogy: From Brownian Motion to Field Fluctuations **5**
2.1 The Mechanism of Giant Fluctuation 5
2.2 Relativistic Selectivity 6
2.3 Conclusion: Order from Chaos 6
2.4 Thermodynamic Argument: Local Condensation vs. Ballistic Fine-Tuning 7

Introduction

The Interpretational Crisis of UHECR Radiation.
Modern high-energy astrophysics finds itself in a state of profound divergence between observation and theory. The detection of particles with energies exceeding 10^{20} eV—such as the famous 1991 "Oh-My-God" event—remains a phenomenon that cannot be fully reconciled with the Standard Model. On one hand, the GZK (Greisen-Zatsepin-Kuzmin) limit imposes an inexorable distance constraint on how far a proton can travel through intergalactic space without losing energy to the Cosmic Microwave Background (CMB). On the other hand, despite decades of searching with the most powerful observatories, such as the Pierre Auger Observatory, we have failed to identify any cosmic accelerators in our local neighborhood capable of imparting such extreme momentum to matter.

This "lack of sources" suggests that the problem lies not within the range of our telescopes, but within the very ontology of how we describe wave-matter interactions. The dominant ballistic paradigm, which treats the photon as a particle carrying discrete momentum, compels us to search for gargantuan, distant magnetic "cannons." However, the formalism of the electromagnetic field energy-momentum tensor ($T^{\mu\nu}$) offers a different, more localized path. If we accept that photon momentum is, in essence, a geometric manifestation of field energy redistribution, the possibility opens for acceleration processes occurring much closer to the observer.

This paper hypothesizes that UHECR particles are the product of local, statistical fluctuations in field energy density during the final phase of their journey toward Earth. Utilizing an analogy to statistical fluid mechanics and Brownian motion, we demonstrate that a relativistic proton is not a passive projectile, but an active element of field dynamics. Under specific geometric conditions, the local field can act as a resonant amplifier, "pumping" energy into the inertial mass of an already speeding particle. Such a framework not only resolves the GZK paradox but redefines the circumterrestrial vacuum as an active, non-linear medium for energy transfer, where extreme events are rare but inevitable statistical fluctuations.

Mieczysław Fabianowicz.
photo: 45 centrum
Prawa zastrzeżone.
Nie wolno przywłaszczyć sobie autorstwa.

1 Astrophysical Implications of the MRG-MLP Model: Resolving the UHECR Paradox

1.1 Critique of the Ballistic Model and the GZK Limit

Classical astrophysics faces a fundamental barrier in explaining the origin of ultra-high-energy cosmic rays (UHECR), such as the *Oh-My-God* particle (10^{20} eV). According to the GZK (Greisen-Zatsepin-Kuzmin) theory, protons at these energies should dissipate energy through interactions with the cosmic microwave background (CMB) over distances shorter than 50 Mpc. The absence of observable sources of such immense power in our local galactic neighborhood suggests a flaw in the underlying assumptions regarding the acceleration mechanism.

In the MLP (Momentum-less Photon) model, we propose that these particles are not "travelers" from distant galaxies, but the result of **local, geometric field acceleration** occurring within the Solar System or the Earth's immediate vicinity.

1.2 The Mechanism of Local Field "Surfing"

Rather than a point-like corpuscular collision, the MLP model describes the process as continuous work performed by an asymmetric field upon the proton's charge. Utilizing the relativistic equation of motion derived in Section A.3:

$$\frac{d}{dt}(\gamma m_p \mathbf{v}) = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

where the driving force arises from the local gradient of the photon field energy (the Poynting vector $\mathbf{S}$).

1.3 The Statistical Singularity of the Event

The rarity of UHECR detection is not a function of cosmic distance, but rather the extremely low geometric probability of a confluence of three factors:

1. **Proton Injection:** The presence of a relativistic "seed" proton (e.g., from the solar wind) within a specific spatial region.

2. **Field Asymmetry:** The emergence of a transient corridor of high field energy density u_{EM} (an asymmetric "photon swarm").

Mieczysław Bartnicewicz.
photon.45@interpl.pl
01.07.2025.
Prawa zastrzeżone.
Nie wolno przepisywać sobie autorstwa.

3. **Directional Coherence:** The proton must move nearly parallel to the energy flux vector $\mathbf{S}$ for a duration on the order of minutes. This allows its charge sphere to "swell," enabling the direct **conversion of field energy into the kinetic energy of mass**. In this process, the particle accumulates the fluctuation potential, transforming it into extreme momentum as measured in the observer's frame.

1.4 Abrupt Decoherence: Explosion of Accumulated Potential

A crucial aspect of the **Energy from Fluctuations** mechanism is its reversibility upon the violation of coherence conditions. When a proton, thus "swollen" with kinetic energy, encounters the dense layers of the Earth's atmosphere, the field absorption process is instantaneously interrupted:

- **Resonator Destruction:** The initial collision with a nucleus (nitrogen, oxygen, or other atmospheric gases) drastically alters the proton's geometry, destroying the relativistic "grating" (Lorentz disk) that facilitated surfing on the fluctuations.
- **Cascading Return to the Field:** The accumulated kinetic energy, which was essentially "trapped" electromagnetic field energy, is violently released. This is not merely a mechanical collision, but a **reconversion**: the mass's kinetic energy reverts to electromagnetic wave form.

This explains the phenomenon of Extensive Air Showers (EAS). The *Oh-My-God* particle does not carry energy "from afar" like a projectile; it brings a locally accumulated field potential into the atmosphere, which "detonates" upon collision, creating a measurable macroscopic energy flash.

1.5 Relativistic Boost Effect

A pivotal element of the local mechanism is the fact that the proton enters the interaction region already possessing high kinetic energy, $E_{\text{init}} \approx 10^{15}$ eV. In the MLP model, this implies that its charge sphere $\rho(x)$ is already in a state of high field tension (a high γ factor).

According to the principle of geometric conversion, the additional energy ΔE acquired from the asymmetry of the Poynting flux $\mathbf{S}$ during the final flight phase (e.g., within the Earth's magnetosphere) sums non-linearly:

$$E_{\text{total}} = \gamma_{\text{init}} m_p c^2 + \int_{t_{\text{entry}}}^{t_{\text{impact}}} \mathbf{F}_{\text{field}} \cdot \mathbf{v}\, dt$$

Mieczysław Sawaczewicz
profesor45@interia.pl
© 2025.
Prawa zastrzeżone.
Nie wolno przypisywać sobie autorstwa.

Because $v \approx c$, the work performed by the photon field is maximally efficient. This allows for a massive energy jump to the level of 10^{20} eV within a time-frame of seconds, without the need to assume unphysically dense external fields. This is the classical equivalent of a Brownian fluctuation occurring within a system already possessing high internal energy.

1.6 Statistical Selectivity of the Acceleration Mechanism

The probability of acquiring additional energy from local photon fields is strongly dependent on the initial relativistic state of the particle. In the MLP model, the probability P of a rapid increase in kinetic energy scales with the Lorentz factor γ:

$$P(\Delta E) \propto \gamma \cdot \exp\left(-\frac{\Delta E}{\Phi_{eff} \cdot \gamma}\right)$$

where Φ_{eff} represents the local interference density of the fields. This implies that the mechanism operates selectively: only particles with initial energies above the "knee" (PeV range) have a non-zero probability of entering the "local surfing" regime. This explains both the observed cosmic ray spectrum profile and the extreme rarity of "Oh-My-God" type events. Acceleration in the final flight phase is, therefore, the result of a geometric resonance between the high-velocity charge and the local structure of the energy-momentum tensor $T^{\mu\nu}$.

Mieczysław Borankiewicz
photon4@mail.com
Prawa zastrzeżone.
Nie wolno przepisywać bez zgody autorstwa.

2 Statistical Analogy: From Brownian Motion to Field Fluctuations

The foundation of the proposed mechanism is a structural isomorphism between classical Brownian motion and the interaction of a relativistic charge with the field noise of the vacuum. In the classical Einstein-Smoluchowski framework, a macroscopic particle (e.g., a micrometer-sized pollen grain) suspended in a liquid exhibits chaotic displacements, even though the statistical sum of collisions with water molecules (H_2O) from all sides should theoretically be zero.

2.1 The Mechanism of Giant Fluctuation

The key to understanding Brownian motion lies in the fact that across extremely short time intervals Δt, the number of collisions n_L (from the

left) and n_R (from the right) is not identical. The instantaneous asymmetry $\Delta n = |n_L - n_R|$ scales as $\sqrt{N}$, where N is the total number of collisions. Although on average $\Delta n/N \to 0$, rare events are inevitable within a statistical distribution—a "streak" of impacts from a single direction providing the pollen grain with sufficient momentum to overcome viscous drag and perform a visible energy jump.

In the MLP model, a PeV-energy proton acts as this "pollen grain," while the local electromagnetic field (photon noise, background radiation, magnetospheric emissions) corresponds to the liquid molecules. From this perspective, the *Oh-My-God* event is nothing more than a giant fluctuation of the Poynting vector $\mathbf{S}$, which, within a narrow temporal and spatial window, "pushes" the proton with extraordinary, asymmetric force.

2.2 Relativistic Selectivity

However, a significant difference in scale exists: in a liquid, the pollen grain is thousands of times larger than the water molecules. In the field model, it is the relativistic boost of the proton (the high γ factor) that determines its susceptibility to fluctuation:

- **Low-energy proton:** It remains "transparent" to minor field asymmetries; the sum of interactions cancels out, yielding no measurable energy gain.
- **Relativistic proton:** Due to Lorentz contraction and Doppler field enhancement, it becomes a "resonator" capable of capturing energy from background fluctuations.

The probability of energy acquisition ΔE by a particle with initial energy E_0 can be described by a modified distribution:

$$P(\Delta E) \propto \exp\left(-\frac{\Delta E}{\xi \cdot \gamma}\right) \tag{1}$$

where ξ represents the fluctuation density parameter of the local $T^{\mu\nu}$ tensor. This equation explains why the cosmic ray spectrum drops precipitously—only an extremely small fraction of particles encounter a "wave" of sufficient amplitude to jump into the UHECR regime.

2.3 Conclusion: Order from Chaos

This analogy redefines the origin of extreme-energy particles. They are not the result of precise "aiming" from distant galactic cannons, but rather

Mieczysław Borancewicz.
proton45@interne.pl
07-07-2025.
Prawa zastrzeżone.
Nie wolno przypisywać osobie autorstwa.

the statistical "peak" of local field activity. Just as Brownian motion proved the molecular structure of matter, the existence of UHECR particles without visible sources serves as the strongest evidence for the dynamic, fluctuating nature of the electromagnetic field as described by the MLP paradigm.

2.4 Thermodynamic Argument: Local Condensation vs. Ballistic Fine-Tuning

The classical "distant source" model requires a non-physical degree of precision, or fine-tuning. For a particle with 10^{20} eV to reach Earth from distances exceeding 50 Mpc, it would have to avoid billions of interactions with CMB photons, which statistically should have decelerated it (the GZK paradox).

In the MLP (Momentum-less Photon) paradigm, the UHECR particle is interpreted as a local "statistical outlier." Rather than assuming the existence of unphysically efficient accelerators in deep space, this model points to a process of **local energy accumulation**. From the perspective of statistical physics, the transfer of energy from field fluctuations to a relativistic resonator (the proton) in our immediate cosmic neighborhood is an order-of-magnitude more probable process than the ballistic transport of energy across the intergalactic void. UHECR is thus not a "messenger from afar," but a local testament to the dynamic, fluctuating nature of the vacuum.

Mieczysław Bałandowicz
protonl5@interia.pl
07-2025
Prawa zastrzeżone
Nie wolno przypisywać sobie autorstwa

Astrophysical Conclusions

The MLP (Momentum-less Photon) model eliminates the need to search for exotic extragalactic sources for "Oh-My-God" particles. It demonstrates that:

- Nuclear stability and momentum are domains of **field geometry**, rather than particle ballistics.
- The cosmic vacuum acts as an active accelerator whenever the field energy flux density exceeds statistical thresholds.
- The "Oh-My-God" particle is a local "surfer" of field energy, which explains the lack of correlation between its arrival direction and persistent astronomical objects.

[

Conclusion: The Field Dynamics Paradigm] Final Conclusion: The Field Dynamics Paradigm

This work proves that eliminating photon momentum not only simplifies the mathematical description of the Compton effect (Section 5.2) but also provides a consistent mechanism for explaining the most enigmatic high-energy phenomena in the Universe. All observed effects—from mass deficits in nuclei to the extreme energies of cosmic radiation—emerge as natural consequences of the inertia of extended charge fields subjected to the velocity barrier c.

Mieczyk

Mieczysław Barancewicz.
proton.45@interia.pl
07-07-2025.
Prawa zastrzeżone:
Nie wolno przypisywać sobie autorstwa.